

Hypothesis Testing

Null Hypothesis (H_0): Default assumption

Alternative Hypothesis (H_1): What we are trying to prove

No. of success/failure
 $p, q \rightarrow$ given
 then use, $\frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$
 else: $\frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$

Two Errors:

- Type-I error: Reject H_0 when H_0 is true (false positive)
- Type-II error: Accept H_0 when H_1 is true (false negative)

Two Metrics:

- T-1 $\{ \alpha = P(\text{Rejection of } H_0 \text{ given that } H_0 \text{ is true}) \{ P(\text{Reject } H_0 | H_0)$
- T-2 $\{ \text{Power of a test } (1 - \beta) \{ P(\text{Accept } H_0 | H_1)$
- $\beta \rightarrow P(\text{acceptance of } H_0 \text{ given that } H_0 \text{ is true})$

Q. Consider one sample $X \sim N(\mu, \sigma)$. Let the null and alternative hypothesis be $H_0: \mu = -3$ and $H_1: \mu = 3$. Consider a test that rejects H_0 if $X > c$ for some constant c .

$\Rightarrow$ What is the significance level of the test?

$H_0 \text{ reject} \rightarrow X > c \text{ (const.)}$

$$\alpha = P(\text{Reject } H_0 | H_0 \text{ is true})$$

$$\alpha = P\left(\frac{X - \mu}{\frac{\sigma}{\sqrt{n}}} > \frac{c - \mu}{\frac{\sigma}{\sqrt{n}}} \mid \mu = -3\right)$$

$$\alpha = P\left(Z > \frac{c + 3}{3}\right) = 1 - P\left(Z < \frac{c + 3}{3}\right)$$

$$1 - F_2\left(\frac{c + 3}{3}\right)$$

$\Rightarrow$ What is the power of the test?

$$\hookrightarrow P(\text{Reject } H_0 | H_1)$$

$$\Rightarrow P(X > c | \mu = 3)$$

$$P\left(\frac{X - \mu}{\frac{\sigma}{\sqrt{n}}} > \frac{c - \mu}{\frac{\sigma}{\sqrt{n}}} \mid \mu = 3\right) = 1 - P\left(Z \leq \frac{c - 3}{3}\right) = 1 - F_2\left(\frac{c - 3}{3}\right)$$

o **Z test** $\rightarrow$ critical values $= \pm 1.96$ (Two-Tailed)
 $\rightarrow$ critical val. $= 1.645$ (Right-Tailed)
 $\rightarrow$ critical val. $= -1.645$ (Left-Tailed)

Q. A survey of 225 Randomly selected students from a city revealed that 89.4% of them have participated in extra curricular activities in their schools. Can we conclude at 1% level of significance that 90% of the students have participated in extra curricular activities?

Sol
 Sample proportion Z-test $\hat{p} \sim N \left(p, \sqrt{\frac{p(1-p)}{n}} \right)$ $\rightarrow$ for large n, the sample proportion behaves like

$$SE = \sqrt{\frac{0.9 \times 0.1}{225}} = 0.02$$

$$Z\text{-score} = \frac{0.894 - 0.90}{0.02} = -0.3$$

Independent 2-sample Test

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\left(\frac{S_1^2}{n_1} \right) + \left(\frac{S_2^2}{n_2} \right)}}$$

Q. The IQs of 25 students from 1 batch of IITM students showed a mean of 110 with a standard deviation of 8, while the IQs of 25 students from another batch of IITM students showed a mean of 115 with a S.D of 7. Is there a significant difference between the IQs of the 2 grps at a 0.05 level of significance?

Significance $t_{2}^{-1}(0.025) = -1.96$

Sol

$$H_0 \Rightarrow \mu_1 = \mu_2$$

$$H_1 \Rightarrow \mu_1 \neq \mu_2 \text{ (2-tailed test)}$$

$$SE = \sqrt{\frac{8^2}{25} + \frac{7^2}{25}} = \frac{\sqrt{113}}{5}$$

$$Z = \frac{110 - 115}{SE} = \frac{-5}{\sqrt{113}/5} \approx -2.35$$

Independent
2-sample
T-test

Decision:

• Critical region: $|z| > 1.96$

• Here: $|z| = 2.35 > 1.96 \rightarrow \text{Ans. Yes}$

f-Test $\rightarrow$ used to compare variability (spread)
b/w 2 grps
 $\swarrow$
Not symmetric

$$f = \frac{\text{variance of grp-2}}{\text{var. of grp-1}} = \frac{S_B^2}{S_A^2}$$

Q. An instructor conducts 2 sessions for 2 diff. classes, A, B.
S-A has 25 students while S-B has 36 students.
Although there was no significant diff. in mean grades,
S-A has a S.D of 10 while S-B had a S.D of 14. Can
we conclude at the 0.01 level of significance that the
variability in marks of class B is greater than that of A?

Sol.

~~H_0~~ $H_0: \sigma_B^2 \leq \sigma_A^2$ $F(35, 24) (0.99) = 2.529$

$H_1: \sigma_B^2 > \sigma_A^2$ (right tailed test)

$$f = \frac{S_B^2}{S_A^2} = \frac{196}{100} = 1.96$$

Critical val. $\rightarrow$ $df_1 = 35$
 $df_2 = 24$

$$F_{0.01}(35, 24) = 2.529$$

Decision: $1.96 < 2.529 \rightarrow$ Fail to reject
 $\downarrow$ $\downarrow$
calc. critical

If its double tailed test, then we need to
Find a upper bound

$$F_{\alpha}(d_1, d_2) = \frac{1}{F_{1-\alpha}(d_2, d_1)}$$

one tail gives 0.2

other give $\frac{1}{0.2} = 5$

o Chi-Square Test (χ^2)

Q. The s.d of weights of 70 gram bags of white cheddar is expected to be 2.5 g. A random sample of 20 packages showed a s.d of 3 grams. Is the apparent increase in variability significant at the 0.05 level?

$$\text{Use } \chi^2_{19}(0.95) = 30.14$$

Sol. $\chi^2 = \frac{(n-1) s^2}{\sigma^2} = \frac{19 \cdot 9}{6.25} = 27.36$

variance testing

Sample size $n = 20$

Sample s.d $s = 3$ ($s^2 = 9$)

Claimed s.d $\sigma = 2.5 \Rightarrow \sigma^2 = 6.25$

o Critical val. $\chi^2_{0.95, 19} = 30.14$

o Decision $\rightarrow$ Calculate $\chi^2 = 27.36$
Critical $\chi^2 = 30.14$

Since, $27.36 < 30.14$,
we fail to reject H_0